Quantum Field Theory Notes

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Contents

1. Math Toolkit

1.1. Notation (Read everything or you won't be able to understand later, or come back to it later)

- (Loose) Einstein Summation notation is used in which if an index appears twice in both top and bottom, such as $\partial_\mu \partial^\mu$, it is equivalent to summing over said indices such as $\sum_{\mu \in \{0,1,2,3\}} \partial_\mu \partial^\mu$
  - The reason that this is loose is due to the fact that if two indices appear twice in an expression but they are both on the bottom, it is implied that the reader should raise the second appearance of the index. Here is an example so you (the reader AND me) don't get confused: $\partial_\mu \partial_\mu = \partial_\mu \partial^\mu = \sum_{\mu \in \{0,1,2,3\}} \partial_\mu \partial^\mu$
- The *Dirac Delta Function* is defined as

$$\delta(x) = \begin{cases} 0 & \text{if } x \neq 0 \\ \infty & \text{if } x = 0 \end{cases} \tag{1}$$

  with

$$\int dx \ \delta(x) = 1 \tag{2}$$

- $\vec{x} = x^i$, $i \in \{1,2,3\}$ (spatial components of x)
- The *d'Alembertian* operator $\square \equiv \partial_\mu \partial^\mu$
- The *Laplacian* operator $\triangle \equiv \partial_i \partial^i$ where $i \in \{1,2,3\}$
- Natural Units are used in which $\hbar = c \ = 1$
- Both Sided Derivative: $\phi^* \overleftrightarrow{\partial_\mu} \phi \equiv \left(\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*\right)$
- Feynman Slash: $\not{p} \equiv p_\mu \gamma^\mu$
- Pauli matrices

$$\sigma^0 = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, \sigma^1 = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}, \sigma^2 = \begin{pmatrix} & -i \\ i & \end{pmatrix}, \sigma^3 = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \tag{3}$$

and $\sigma^\mu \equiv (\sigma^0, \vec{\sigma})$, $\bar{\sigma}^\mu \equiv (\sigma^0, -\vec{\sigma})$

- $$\gamma^\mu = \begin{pmatrix} & \sigma^\mu \\ \bar{\sigma}^\mu & \end{pmatrix} \tag{4}$$

  in the Weyl representation
- $\bar{\psi} \equiv \psi^\dagger \gamma^0$ where ψ is a Dirac spinor
- Left Handed Derivative

$$\begin{aligned} &\bar{\psi}\left(-i\overleftarrow{\not{\partial}} - m\right) \\ &= -i\partial_\mu \bar{\psi} \gamma^\mu - m\bar{\psi} \end{aligned} \tag{5}$$

  (Derivative acts on the left)
- $k_\mu^2 \equiv k_\mu k^\mu$
- ψ^α left handed Weyl spinor

- $\tilde{\psi}_{\dot{\alpha}}$ right handed Weyl spinor
- $\psi = \begin{pmatrix} \psi^{\alpha} \\ \tilde{\chi}_{\dot{\beta}} \end{pmatrix}$ Dirac spinor (Representation of $\left(\frac{1}{2}, 0\right) \oplus \left(0, \frac{1}{2}\right)$)

1.2. Integrals that are useful

- $$\int \mathrm{d}x \ f(x)\delta(x^2 - a^2) = \frac{1}{2|a|}(f(a) + f(-a)) \tag{6}$$
- $$\int \mathrm{d}^3 k \ \theta(k^0)\delta\left(k_\mu^2 - m^2\right) = \frac{1}{2\omega_k}[\theta(\omega_k) + \theta(-\omega_k)] = \frac{1}{2\omega_k} \tag{7}$$

  where $\omega_k = \sqrt{\vec{k}^2 - m^2}$
- $$\int \frac{\mathrm{d}^4 k}{(2\pi)^4} \exp(ik(x-y)) = \delta^4(x-y) \tag{8}$$
- $$\int \frac{\mathrm{d}^N k}{(2\pi)^N} \exp(ik(x-y)) = \delta^N(x-y) \tag{9}$$
- $$\int \mathrm{d}x \exp\left(-\frac{1}{2}ax^2 + Jx\right) = \left(\frac{2\pi}{a}\right)^{\frac{1}{2}} \exp\left(\frac{J^2}{2a}\right) \tag{10}$$

1.3. Group Theory

Definition 1.3.1: A Group G is a set of elements $\{g_i\} \in G$ and a rule $g_i \times g_j = g_k$ giving how they "multiply."

Group Theory Axioms

- For each element $g_i \in G$ and $g_j \in G$, their product $g_i \times g_j = g_k \in G$ is an element of G
- Products must be Associative:

$$\left(g_i \times g_j\right) \times g_k = g_i \times \left(g_j \times g_k\right) \tag{11}$$

- There must be an Identity:

$$\mathbb{1} \times g_i = g_i \times \mathbb{1} = g_i \tag{12}$$

- Elements have an Inverse:

$$g_i^{-1} \times g_i = \mathbb{1} \tag{13}$$

Definition 1.3.2 (Representation): An embedding of g_i as operators in a vector space

- Matrices are finite-dimensional representations of a group

Definition 1.3.3 (Faithful Representation): Each element in a group gets its own matrix

1.3.1. Lie Groups

- A Group G whose elements can be written as $g_i = \exp(i c_i^g \lambda_i)$ for some number c_i^g and some matrix λ_i (known as a *generator*)
- In an algebra, you can add and multiply
- In a Group, you have to follow the previously mentioned rules (just multiplication is allowed)

Definition 1.3.1.1 (Lie Group): A *Lie Group* is a type of group with an infinite number of elements and a finite number of generators.

Definition 1.3.1.2 (Lie Algebra): The Generators of a *Lie group* form a *Lie Algebra*.

Example 1.3.1.1 (Lie Algebras):
- QED : $\mathrm{U}(1)$ (unitarity)
- Weak: $\mathrm{SU}(2)$ (special unitarity)
- Strong: $\mathrm{SU}(3)$
- Lorent Group: $\mathrm{O}(1,3)$ (orthogonal and preserving (1,3) metric) (see §2.2)

2. Lorentz and Spinors

2.1. Lorentz Transformations

Definition 2.1.1 (Metric Tensor): A (or in special relativity, *The*) metric tensor is defined as being the dot product of basis vectors

$$g_{\mu\nu} = \hat{e}_{(\mu)} \cdot \hat{e}_{(\nu)} \tag{14}$$

This definition is all but useless in Quantum Field Theory due to the whole theory entirely using Special Relativity instead of General Relativity[1]

The special relativity metric tensor (which is subject to arguments over convention), is either $g_{\mu\nu} = \text{diag}(+,-,-,-)$ or $g_{\mu\nu} = \text{diag}(-,+,+,+)$[2].

Definition 2.1.2 (Lorentz Transformation & Lorentz Invariance): A *Lorentz transformation* Λ is defined as a coordinate transformation

$$\Lambda : (t, x, y, z) \to (t', x', y', z') \tag{15}$$

such that

$$\Lambda^T g \Lambda = g \tag{16}$$

or

$$\Lambda^\mu_\alpha \Lambda^\nu_\beta \eta_{\mu\nu} = \eta_{\alpha\beta} \tag{17}$$

In special relativity (minkowski space), this condition is loosened to

$$s^2 = t^2 - x^2 \tag{18}$$

is conserved but the previous condition is still useful in discussion of group theory and derivation of *Spinors*. Generally, Lorentz transformations are transformations that

2.2. Group Theory (cont'd)

Definition 2.2.1 (Lorentz Group): This is the set of operations Λ (in the 4-vector representation) such that

$$\Lambda^T g \Lambda = g \tag{19}$$

- 4-vector representation

[1]Unless string theory, again
[2]Personally I am used to $\text{diag}(+,-,-,-)$.

$$x_\mu \to \Lambda_{\mu\nu} x_\nu \tag{20}$$

- Lorentz transformations are *rotations* and *boosts*
- Rotations:
- Can utilize *infinitesimal* Lorentz transformations in order to extract the group ($O(1,3)$) (we let β_i parametrize boosts and θ_i parametrize rotations)
  - $$\begin{aligned} \delta x_0 &= \beta_i x_i \\ \delta x_i &= \beta_i x_o - \varepsilon_{ijk} \theta_j x_k \end{aligned} \tag{21}$$

  or

  $$\delta x_\mu = i \left[\sum_{i \in \{1,2,3\}} \theta (J_i)_{\mu\nu} + \beta_i (K_i)_{\mu\nu} \right] x_\nu \tag{22}$$

  - J_i and K_i matrices:

  $$\begin{gathered} J_1 = i \begin{pmatrix} 0 & & & \\ & 0 & & \\ & & 0 & -1 \\ & & 1 & 0 \end{pmatrix}, J_2 = i \begin{pmatrix} 0 & & & \\ & 0 & & 1 \\ & & 0 & \\ & -1 & & 0 \end{pmatrix}, J_3 = i \begin{pmatrix} 0 & & & \\ & 0 & -1 & \\ & 1 & 0 & \\ & & & 0 \end{pmatrix} \\ K_1 = i \begin{pmatrix} 0 & -1 & & \\ -1 & 0 & & \\ & & 0 & \\ & & & 0 \end{pmatrix}, K_2 = i \begin{pmatrix} 0 & & -1 & \\ & 0 & & \\ -1 & & 0 & \\ & & & 0 \end{pmatrix}, K_3 = i \begin{pmatrix} 0 & & & -1 \\ & 0 & & \\ & & 0 & \\ -1 & & & 0 \end{pmatrix} \end{gathered} \tag{23}$$

  - Such matrices are *generators of the Lorentz group*
    - Can use to obtain the Lie Algebra
  - $$\Lambda = \exp(i \theta_i J_i + i \beta_i K_i) \tag{24}$$

2.3. Spinors

TODO

3. Lagrangians

Definition 3.1 (Lagrangian): The *Lagrangian* (Density) $\mathcal{L}$, similar to the *Hamiltonian* H (or $\mathcal{H}$ for Hamiltonian density, defined as $H = \int \mathrm{d}^3x\mathcal{H}$) as they both can be used to describe the evolution of a system. In Field Theories, densities are used more commonly. The Density of either quantity, as stated before, is a more useful quantity in Field theories due to Lorentz Invariance and other such things.

Definition 3.2 (Action): Action S is defined as

$$S = \int \mathrm{d}tL = \int \mathrm{d}^4x\mathcal{L} \tag{25}$$

which should be familiar from studying classical mechanics (Lagrangian mechanics specifically). This quantity is minimized in order to obtain the evolution of a system (**Equations of Motion**).

Theorem 3.1 (Euler Lagrange Equations): In order to minimize the action of a system with Lagrangian Density $\mathcal{L}[\phi, \partial_\mu\phi]$, the system must satisfy the ***Euler Lagrange Equations***:

$$\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = 0 \tag{26}$$

A differential equation which ends up being extremely useful in physics for describing the time evolution of a system.

Proof: We aim to equate the variation of the action S with respect to the field ϕ to zero. That is, we require

$$\frac{\delta S}{\delta\phi} = 0 \tag{27}$$

Note that

$$S = \int \mathrm{d}tL = \int \mathrm{d}^4x\mathcal{L}[\phi, \partial_\mu\phi] \tag{28}$$

in order to obtain δS, we must perform a transformation $\phi \to \phi + \delta\phi$ in the Lagrangian. This provides

$$\delta S = \int d^4x\left(\frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta(\partial_\mu\phi)\right) \tag{29}$$

Via the product rule and the fact that infinitesimal variations commute with partial derivatives[3], we have that

$$\begin{aligned}
\delta S &= \int \mathrm{d}^4x \left(\frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi \right) - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi \right) \\
\delta S &= \int \mathrm{d}^4x \left(\frac{\partial \mathcal{L}}{\partial \phi}\delta\phi - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi \right) \right) \\
\delta S &= \int \mathrm{d}^4x \left(\left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right)\delta\phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi \right) \right)
\end{aligned} \tag{30}$$

Now we need to prove that the final term (the one without the shared common factor of $\delta\phi$) vanishes as our integration bounds become infinite (as we are integrating over all spacetime of course). Since this term is a total derivative of some quantity that we will define to be

$$A^\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi \tag{31}$$

(Something similar with the form of $\frac{A^\mu}{\delta\alpha}$ is a physically important quantity but this A^μ is not of any significance besides being useful for math). With the derivative being of course $\partial_\mu A^\mu$. The integral of this quantity is

$$\int \mathrm{d}^4x \partial_\mu A^\mu \tag{32}$$

and by the **4 dimensional divergence theorem**, we obtain

$$\int_V \mathrm{d}^4x \partial_\mu A^\mu = \oint_{\partial V} \mathrm{d}^3\sigma_\mu A^\mu \tag{33}$$

This must vanish, given that the $\delta\phi$ is forced to vanish at the boundaries by the principle of least action. Therefore, we have that

$$\frac{\delta S}{\delta \phi} = \int \mathrm{d}^4x \left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = 0 \tag{34}$$

For the integral to vanish, the integrand must vanish, therefore

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = 0} \tag{35}$$

■

[3]Look into variational calculus if you want to learn more about this

Theorem 3.2 (Extended Euler Lagrange Equations): In order to minimize the action of a system with Lagrangian Density $\mathcal{L}[\phi, \partial_\mu\phi, \partial_\nu\partial_\mu\phi, ...]$, the system must satisfy the ***Extended Euler Lagrange Equations***:

$$\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} + \partial_\nu\partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\nu\partial_\mu\phi)} - \cdots = 0 \tag{36}$$

(This was actually a problem in the Schwartz QFT textbook, so if you are reading that, potentially skip this part.)

Proof: We begin in a similar way to the proof to the *Euler-Lagrange equations.*

$$\delta S = \int \mathrm{d}^4x \left(\frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta(\partial_\mu\phi) + \frac{\partial\mathcal{L}}{\partial(\partial_\nu\partial_\mu\phi)}\delta(\partial_\nu\partial_\mu\phi) + \cdots \right) \tag{37}$$

We then utilize the product rule (or the reverse of it) to obtain

$$\begin{aligned} \partial_\nu\partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\nu\partial_\mu\phi)}\delta\phi \right) = {} & \partial_\nu\partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\nu\partial_\mu\phi)}\delta\phi \\ & + \partial_\nu \frac{\partial\mathcal{L}}{\partial(\partial_\nu\partial_\mu\phi)}\delta(\partial_\mu\phi) \\ & + \partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\nu\partial_\mu\phi)}\delta(\partial_\nu\phi) \\ & + \frac{\partial\mathcal{L}}{\partial(\partial_\nu\partial_\mu\phi)}\delta(\partial_\nu\partial_\mu\phi) \end{aligned} \tag{38}$$

We can rewrite $\partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\nu\partial_\mu\phi)}\delta(\partial_\nu\phi)$ and $\partial_\nu \frac{\partial\mathcal{L}}{\partial(\partial_\nu\partial_\mu\phi)}\delta(\partial_\mu\phi)$ via similar methods and we can continue to do so with more terms forming a pattern of the nth (with n partial derivatives) term has a coefficient of $(-1)^n$ (this method similar to the *principle of inclusion-exclusion*)

$$\begin{aligned} \delta S = \int \mathrm{d}^4x \Bigg(& \left(\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} + \partial_\nu\partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\nu\partial_\mu\phi)} - \cdots \right)\delta\phi \\ & + \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi \right) + \partial_\mu \left(\partial_\nu \frac{\partial\mathcal{L}}{\partial(\partial_\nu\partial_\mu\phi)}\delta\phi \right) + \partial_\nu \left(\partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\nu\partial_\mu)}\delta\phi \right) \\ & + \partial_\mu\partial_\nu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\partial_\nu)}\delta\phi \right) + \cdots \Bigg) \end{aligned} \tag{39}$$

Every term besides those with the shared factor of $\delta\phi$ disappears due to similar reasons mentioned in the proof of the *Euler-Lagrange equations* (i.e. they are total derivatives). and therefore, since we want $\frac{\delta S}{\delta\phi} = 0$, we have that

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} + \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} - \cdots = 0} \tag{40}$$

■

Example 3.1 (Lagrangian): An example of a Lagrangian that can apply to some general scalar field ϕ[4] is the Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial_\mu \phi) - V[\phi] \tag{41}$$

This would give us the equation of motion

$$\begin{gathered} \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = 0 \\ -V'[\phi] - \Box\phi = 0 \end{gathered} \tag{42}$$

so

$$(\Box + V'[\phi])\phi = 0 \tag{43}$$

which is related to the *Klein-Gordon equation* $(\Box + m^2)\phi = 0$.

- One can split the terms of a Lagrangian $\mathcal{L}$ into *Kinetic* terms and *Interacting* terms.
  - *Kinetic* terms of a Lagrangian are ones which have at most 2 fields in it
    -
$$\mathcal{L}_{\text{kin}} \supset \frac{1}{2}\phi\Box\phi, \bar{\psi}\not{\partial}\psi, \frac{1}{4}F_{\mu\nu}^2, \frac{1}{2}m^2\phi^2, \cdots \tag{44}$$

- Interacting terms are terms with more than 2 fields

$$\mathcal{L}_{\text{int}} \supset \lambda\phi^3, g\bar{\psi}\not{A}\psi, g\partial_\mu\phi A_\mu \phi^\star, \cdots \tag{45}$$

> **Note**: Interacting terms in the lagrangian all have a factor λ or g associated with them. This is due to the fact that it is a coupling constant related to the interaction.

[4]That is, it doesn't apply universally, but just in this circumstance

4. Classical Symmetries

4.1. Noether's Theorem

- This theorem is the core in the utilizing symmetries in classical and non-classical field theory.

Theorem 4.1.1 (Noether's Theorem): If a Lagrangian has a continuous symmetry (i.e. it is parameterizable by some continuous α), there exists a conserved current J^μ associated with said symmetry when the equations of motion of the Lagrangian are satisfied, i.e. the **Euler-Lagrange** equations. In other words, *symmetries imply conservation laws.*

Proof: The Proof for *Noether's Theorem* is fairly short and clever.

We define $\mathcal{L}$ as the Lagrangian of the system, ϕ_n as the field(s) involved in the system, and α as the parameter of the continuous symmetry.

Similar to the derivation of the *Euler-Lagrange* equations,

$$0 = \frac{\delta \mathcal{L}}{\delta \alpha} = \sum_n \left(\left(\frac{\partial \mathcal{L}}{\partial \phi_n} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \right) \frac{\delta \phi_n}{\delta \alpha} + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} \right) \tag{46}$$

This expression must be equivalent to zero due to α parametrizing a symmetry (i.e. the Lagrangian is invariant wrt some variation in the parameter). It is necessary to define a

$$J^\mu \equiv \sum_n \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} \tag{47}$$

Since we are only considering the case in which the equations of motion hold true, (i.e. the *Euler-Lagrange equations* are true), we have that

$$\boxed{\partial_\mu J^\mu = \partial_\mu \sum_n \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} = \sum_n \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} = 0} \tag{48}$$

Therefore due to some continuous symmetry parametrized by some α, there exists a conserved current J^μ. ∎

- The *Energy-Momentum Tensor* $\mathcal{T}_{\mu\nu}$ is the conserved current for spacetime translation.

Definition 4.1.1 (Energy Momentum Tensor): The *Energy-Momentum Tensor* $\mathcal{T}_{\mu\nu}$ for some Lagrangian $\mathcal{L}[\phi_1, \partial_\mu \phi_1, ... \phi_n, \partial_\mu \phi_n]$ and some spacetime translation $x^\nu \to x^\nu + \xi^\nu$ is defined as

$$\mathcal{T}_{\mu\nu} \equiv \sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu} \mathcal{L} \tag{49}$$

Derivation below.

Starting with the translation

$$\phi(x) \to \phi(x+\xi) = \phi(x) + \partial_\nu \xi^\nu \phi(x) + \dots \tag{50}$$

This provides $\frac{\delta\phi}{\delta\xi^\nu} = \partial_\nu \phi$ which applies universally regardless of the field ϕ (and in fact regardless of the physical object, as it applies to the Lagrangian $\mathcal{L}$, a consequence of the fact that $\delta S = 0$).

We then have that

$$\begin{aligned}
\frac{\delta\mathcal{L}}{\delta\xi^\nu} &= \partial_\mu \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi_n)} \frac{\delta\phi_n}{\delta\xi^\nu} \right) \\
&= \partial_\mu \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n \right) \\
\partial_\nu \mathcal{L} &= \partial_\mu \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n \right)
\end{aligned} \tag{51}$$

We then have that

$$\partial_\mu \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu}\mathcal{L} \right) = 0 \tag{52}$$

Where we then obtain

$$\mathcal{T}_{\mu\nu} \equiv \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu}\mathcal{L} \right) \tag{53}$$

and $\partial_\mu \mathcal{T}_{\mu\nu} = 0$.

- Energy density:

$$\mathcal{E} = \mathcal{T}_{00} = \sum_n \frac{\partial\mathcal{L}}{\partial(\dot{\phi}_n)} \dot{\phi} - \mathcal{L} \tag{54}$$

  - $\dot{\phi}_n = \partial_t \phi_n$
  - In the case of Lagrangians with $\dot{\phi} = \pi = \frac{\partial\mathcal{L}}{\partial\phi}$, $\mathcal{E} = \mathcal{H}$ due to the *Legendre Transform*

Definition 4.1.2 (4-Momentum): In terms of the energy-momentum tensor, 4-momentum p^μ is defined as

$$p^\nu \equiv \int \mathrm{d}^3 x \mathcal{T}^{0\nu} = (E, \vec{p}) \tag{55}$$

4.1.1. More information on currents

- A current can refer to a Noether current (such as the ones previously seen)
- They can be external, such as a moving electron in a wire
- They can be sources for fields

- ▸ They are never fields with kinetic terms
- They can be placeholders in Lagrangians

Example 4.1.1.1 (Currents in Lagrangians): In

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \tilde{\phi} \Box \phi - ie A_\mu \left(\tilde{\phi} \overleftrightarrow{\partial}_\mu \phi \right), \tag{56}$$

we can write it as

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \tilde{\phi} \Box \phi - A_\mu J^\mu \tag{57}$$

- Clearly, you can see how currents are used to represent quantities in lagrangians that we do not care for.

5. Quantization

5.1. Harmonic Oscillator/First Quantization

- The standard Hamiltonian for a simple harmonic oscillator is

$$H = \frac{1}{2}\frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 \tag{58}$$

Definition 5.1.1 (First Quantization): The idea behind first quantization is to apply the commutation relation $[\hat{x}, \hat{p}] = i$ to a system.

- Utilizing a change of variables to a and $a^\dagger$, one can derive an alternative form for the Hamiltonian
  - $$a \equiv \sqrt{\frac{m\omega}{2}}\left(x + \frac{ip}{m\omega}\right),\ a^\dagger \equiv \sqrt{\frac{m\omega}{2}}\left(x - \frac{ip}{m\omega}\right) \tag{59}$$
  - This implies that

$$H = \omega\left(a^\dagger a + \frac{1}{2}\right) \tag{60}$$

  - The eigenstates of this Hamiltonian are eigenstates of $\hat{N} \equiv a^\dagger a$
- $\hat{N}$ operator:
  - $\hat{N}|n\rangle = n|n\rangle$
  - $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$
  - $a|n\rangle = \sqrt{n}|n-1\rangle$

5.1.1. Heisenberg Picture

- $$i\frac{\mathrm{d}}{\mathrm{d}t}a = [a, H] = \left[a, a\left(a^\dagger a + \frac{1}{2}\right)\right] = \omega\left(aa^\dagger a - a^\dagger aa\right) = \omega\left[a, a^\dagger\right]a = \omega a \tag{61}$$
- Solution is

$$a(t) = e^{-i\omega t}a(0) \tag{62}$$

5.1.2. Relativistic Fields

- A common relativistic equation (heavy foreshadowing) for a (free) field is the equation

$$\Box\phi = 0. \tag{63}$$

  - Solution to this is

$$\phi(x, t) = \int \frac{\mathrm{d}^3 p}{(2\pi)^3}\left[a_p(t)e^{i\vec{p}\cdot\vec{x}} + \tilde{a_p}(t)e^{-i\vec{p}\cdot\vec{x}}\right] \tag{64}$$

  which is written in a fairly suggestive manner

5.2. 2nd Quantization

Definition 5.2.1 (Second Quantization): To obtain 2nd quantization (and hence quantum field theory), one can use an annihilation operator a_p and a creation operator $a_p^\dagger$ which exist per wave number as opposed to being universal since now, we utilize infinite harmonic oscillators which exist per p

- Free Hamiltonian is obtained via integration over these

$$H = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \omega_p \left(a_p^\dagger a_p + \frac{1}{2} \right) \tag{65}$$

  with $\omega_p = |\vec{p}|$
- 2nd Quantization new things:
  - Each (of the infinitely many) point of momentum $\vec{p}$ gets its own harmonic oscillator (which we integrate over to obtain our result)
  - The nth excitation of the harmonic oscillator $\vec{p}$ as being a state with n particles

5.2.1. Fock Space

- Instead of Hilbert space $\mathcal{H}$, like in Quantum Mechanics/First Quantization, we have a *Fock Space* $\mathcal{F}$.
- A Fock space $\mathcal{F}$ is the direct sum of Hilbert spaces $\mathcal{H}_n$, that is,

$$\mathcal{F} = \bigoplus_n \mathcal{H}_n \tag{66}$$

- States in $\mathcal{H}_n$ are linear combinations of $\left\{ \left| p_1^\mu ... p_n^\mu \right\rangle \right\}$
  - Since we are using *Fock Space*, this means that there can be however many particles due to the direct sum over Hilbert Spaces

Note (Momentum Notation): $p^0 = \sqrt{\vec{p}^2 + m^2}$ so $|p\rangle = |\vec{p}\rangle = |p^\mu\rangle$ due to no loss of generality (this only works if it is on-shell).

5.2.2. Field Expansion

- Previously, we noted that $\left[a, a^\dagger\right] = 1$. This generalizes to $\left[a_k, a_p^\dagger\right] = (2\pi)^3 \delta^3 \left(\vec{p} - \vec{k}\right)$
- $a_p^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_p}} |p\rangle$
- $\langle 0|0\rangle = 1$ since it is the vacuum ground state
  - $\langle p|k\rangle = 2\sqrt{\omega_p \omega_k} \langle 0| a_p a_o^\dagger |0\rangle = 2\omega_p (2\pi)^3 \delta^3 \left(\vec{p} - \vec{k}\right)$
- Identity operator

$$\mathbb{1} = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \frac{1}{2\omega_p} |p\rangle \langle p| \tag{67}$$

- Free field definition

$$\phi_0(\vec{x}) \equiv \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left(a_p e^{i\vec{p}\cdot\vec{x}} + a_p^\dagger e^{-i\vec{p}\cdot\vec{x}}\right) \tag{68}$$

which is why I said $\phi(\vec{x}, t)$ from 4.1.2 was written in a suggestive manner (although this is still technically just a definition)

Note (Momentum state inner product with free field):

$$\begin{aligned}
\langle\vec{p}|\phi_0(\vec{x})|0\rangle &= \langle 0|\sqrt{2\omega_p} a_p \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \left(a_k e^{i\vec{k}\vec{x}} + a_k^\dagger e^{-i\vec{k}\cdot\vec{x}}\right)|0\rangle \\
&= \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \sqrt{\frac{\omega_p}{\omega_k}} \left[e^{i\vec{k}\cdot\vec{x}} \langle 0|a_p a_k|0\rangle + e^{-i\vec{k}\cdot\vec{x}} \left\langle 0\middle|a_p a_k^\dagger\middle|0\right\rangle = e^{-i\vec{p}\cdot\vec{x}}\right] \\
&= e^{-i\vec{p}\cdot\vec{x}}
\end{aligned} \tag{69}$$

This is the expected result of $\langle\vec{p}|\vec{x}\rangle$, and so we conclude $\phi_0(\vec{x})|0\rangle = |\vec{x}\rangle$

5.3. Time Dependence

- Because of Heisenberg's equations of motion, in the Heisenberg perspective, $a_p(t)$ and $a_p^\dagger(t)$ are time dependent
  - $a_p(t) = e^{-i\omega_p t} a_p, a_p^\dagger(t) = e^{i\omega_p t} a_p^\dagger$
- Heisenberg equations of motion
  - $$\begin{aligned}
[H_0, \phi_0(\vec{x}, t)] &= \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left[\omega_p \left(a^\dagger a_p + \frac{1}{2}\right), a_k e^{-ikx} + a_k^\dagger e^{ikx}\right] \\
&= \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left[-\omega_p a_p e^{-ipx} + \omega_p a_p^\dagger e^{ipx}\right] \\
&= -i\partial_t \phi_0(\vec{x}, t)
\end{aligned} \tag{70}$$
  - Should have that, for any field ϕ,

$$[H, \phi(\vec{x}, t)] = i\partial_t \phi(\vec{x}, t) \tag{71}$$

    - $$\phi(\vec{x}, t) = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left[a_p(t) e^{-ipx} + a_p^\dagger(t) e^{ipx}\right] \tag{72}$$

    is the solution to the *Heisenberg equations of motion* for ϕ, our field which does not necessarily have to be free in this case

6. Cross Sections

Important Result of this Section

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\text{CM}} = \frac{1}{64\pi^2 E_{\text{CM}}^2}|\mathcal{M}|^2 \tag{73}$$

and the Born approximation

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\text{Born}} = \frac{m_e^2}{4\pi^2}\left[\tilde{V}\left(\vec{k}\right)\right]^2 \tag{74}$$

- Cross sections used to make sense of particle accelerator data
- Cross sectional area:

$$\begin{aligned}\sigma &= \frac{\#\text{ particles}}{\text{time} \times \text{number density} \times \text{velocity of beam}} \\ &= \frac{1}{T}\frac{1}{\Phi}N, \Phi = \#\text{ density} \times \text{beam vel.}\end{aligned} \tag{75}$$

  - More useful/natural is differential cross section

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \tag{76}$$

    for some angle/change in angle Ω.
  - Info about shape of object with this
- QM
  - Generalize to Cross Sections
  - In QM, there is a probability of scattering

$$P = \frac{N}{N_{\text{inc}}} \tag{77}$$

    with N_{inc} number of incident particles
  - Cross section

$$d\sigma = \frac{1}{T}\frac{1}{\Phi}dP \tag{78}$$

    (normalize flux)
  - Differential scattering number

$$\mathrm{d}N = L \times \mathrm{d}\sigma \tag{79}$$

7. Propagators

7.1. Green's Functions

Definition 7.1.1 (Green's Functions): A Green's function is a function of the form A^{-1} where A is an operator. A standard way to evaluate these green function operators is via a Fourier transform into k-space[5]. This is a type of *propagator*.

Definition 7.1.2 (Propagator): Defines or is used to define how a field propagates via applying the propagator Π to some other field or current. Can be used perturbatively in order to describe more complicated fields.

- An example of a propagator is the 2-point Green's function $\Pi = -\frac{1}{\square}$ which is defined by

$$\square_x \Pi(x, y) = -\delta^4(x - y) \tag{80}$$

- With $\square_x = g^{\mu\nu} \frac{\partial}{\partial x^\mu} \frac{\partial}{\partial x^\nu}$ This gives the solution (via the Fourier transform), that

$$\Pi(x, y) = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} e^{ik(x-y)} \frac{1}{k^2} \tag{81}$$

.

Note (Fourier Transform of Operators): Under *Fourier Transforms*, certain operators are effectively interchangeable with some counterpart that often simplifies calculations. An example of this is how $\Pi(x, y)$ is interchangeable with $\frac{1}{k^2}$.

[5] k-space is effectively momentum space due to the relation $p = \hbar k$ and the fact that we are using natural units

Example 7.1.1 (Delta Functions): For an instance of this occurrence, we can utilize delta functions.

> **Note**: The Fourier transform of any delta function is
>
> $$\begin{gathered}\tilde{\delta}\left(\vec{k}\right) = 1 \\ \Longrightarrow \delta(\vec{x}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}}\end{gathered} \tag{82}$$

- $$\begin{aligned}\triangle^n\, \delta^3(\vec{x}) &= \int \frac{\mathrm{d}^3 k}{(2\pi)^3}\, \triangle^n\, e^{i\vec{k}\cdot\vec{x}} \\ &= \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \left(-\vec{k}^2\right)^n e^{i\vec{k}\cdot\vec{x}}\end{aligned} \tag{83}$$
- $\triangle^n \longleftrightarrow \left(-\vec{k}^2\right)^n$ (this means that in a *Fourier Transform*, these are interchangeable).
- Additionally, we can do the same for the relativistic $\Box = \partial_\mu \partial^\mu$

$$\Box^n\left(\delta^4(x)\right) = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \Box^n e^{ik\cdot x} = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \left(-k^2\right)^n e^{ik\cdot x} \tag{84}$$

- Have that $[\widetilde{\Box^n \delta^4}](k) = \left(-k^2\right)^n$ and $[\widetilde{\triangle^n\, \delta^3}](k) = \left(-\vec{k}^2\right)^n$

7.2. Perturbation Theory

Definition 7.2.1 (Perturbation Theory): This topic is introduced in Quantum mechanics (and often before-hand) where we consider an initial state that we can solve and then higher order terms that we continue to be able to solve. Perturbation theory only works when the leading term is the "dominant" force in play. Often, we can utilize an initial term which is *non-interacting* and extra terms that are *interacting*.

Example 7.2.1 (Gravity): This example was provided from the Schwartz textbook[6] We start with the Lagrangian

$$\mathcal{L} = -\frac{1}{2}h\Box h + \frac{1}{3}\lambda h^3 + Jh \tag{85}$$

where J is some *current* and h is our field. From the *Euler-Lagrange equations*, our equation of motion is

$$\Box h - \lambda h^2 - J = 0 \tag{86}$$

First order approx ($\lambda = 0$): $h_0 = \frac{1}{\Box}J$. We let $h = h_0 + h_1$.

$$\Box(h_0 + h_1) - \lambda(h_0 + h_1)^2 - J = 0 \tag{87}$$

implies

$$\begin{gathered} \Box h_1 = \lambda h_0^2 + \mathcal{O}(\lambda^2) \\ h = \frac{1}{\Box}J + \lambda\frac{1}{\Box}\left(\left(\frac{1}{\Box}J\right)\left(\frac{1}{\Box}J\right)\right) + \mathcal{O}(\lambda^2) \end{gathered} \tag{88}$$

which we now have in terms of our Green's Functions.

Note: In fact, something being defined as $\frac{1}{\Box}J$ is not that uncommon, as it also happens with the Electromagnetic field.

7.3. OFPT

- For **Old Fashioned Perturbation Theory** we utilize perturbation by setting the Hamiltonian to

$$H = H_0 + V \tag{89}$$

- The idea is that we have some $|\phi\rangle$ and some $|\psi\rangle$ such that

$$H_0|\phi\rangle = E|\phi\rangle \tag{90}$$

and

$$H|\psi\rangle = E|\psi\rangle \tag{91}$$

- From this we can derive

$$|\psi\rangle = |\phi\rangle + \frac{1}{E - H_0}V|\phi\rangle, \tag{92}$$

the *Lippmann-Schwinger* equation
  - This allows us to define the propagator[7]

[6]This is the primary textbook that I am learning Quantum Field Theory through, as seen in the credits section.

[7]In fact, this is not well-defined, but the limit

$$\Pi_{\mathrm{LS}} \equiv \frac{1}{E - H_0} \tag{94}$$

- We define an operator T as

$$\begin{aligned} V|\psi\rangle &= T|\phi\rangle \\ |\psi\rangle &= |\phi\rangle + \Pi_{\mathrm{LS}} T|\phi\rangle \\ V|\psi\rangle &= V|\phi\rangle + V\Pi_{\mathrm{LS}} T|\phi\rangle \end{aligned} \tag{95}$$

  - Note that $V|\psi\rangle = T|\phi\rangle$ so

$$T|\phi\rangle = V|\phi\rangle + V\Pi_{\mathrm{LS}}T|\phi\rangle = (V + V\Pi_{\mathrm{LS}}T)|\phi\rangle. \tag{96}$$

  (This holds when contracted with any eigenstate of H_0, $|\phi_j\rangle$)

- This implies that that $T = V + V\Pi_{\mathrm{LS}}T$.
- Recursion provides

$$T = V + V\Pi_{\mathrm{LS}}V + V\Pi_{\mathrm{LS}}V\Pi_{\mathrm{LS}}V + \cdots \tag{97}$$

If we want a transition amplitude for this, we want to calculate

$$\langle f|T|i\rangle = \langle f|V|i\rangle + \langle f|V\Pi_{\mathrm{LS}}V|i\rangle + \cdots \tag{98}$$

> **Note**: We can insert the eigenstates within this equation due to the fact that
>
> $$\sum_j |\phi_j\rangle\langle\phi_j| = \mathbb{1} \tag{99}$$

This provides

$$\langle\phi_f|T|\phi_i\rangle = \langle\phi_f|V|\phi_i\rangle + \langle\phi_f|V\Pi_{\mathrm{LS}}|\phi_j\rangle\langle\phi_j|V|\phi_i\rangle + \cdots \tag{100}$$

- In an alternative/more convenient notation,

$$T_{fi} = V_{fi} + \left(\sum_j\right) V_{fj}\Pi_{\mathrm{LS}}(j)V_{ji} + \left(\sum_j\right) V_{fi}\Pi_{\mathrm{LS}}(j)V_{jk}\Pi_{\mathrm{LS}}(k)V_{ki} + \cdots \tag{101}$$

  Where $T_{ij} = \langle\phi_i|T|\phi_j\rangle$, $V_{ij} = \langle\phi_i|V|\phi_j\rangle$, $\Pi_{\mathrm{LS}}(j) = \frac{1}{E-E_j}$ with $E_f = E_i$ and the sums tend to be omitted

> **Note** (OFPT With particle interactions): We can with a certain particle state $|i\rangle$ and end with a final certain particle state $|f\rangle$ and use *OFPT* to calculate what happens in-between such states. An example of this is the case of an electron scattering off an electron.

$$\lim_{\varepsilon \to 0} \frac{1}{E - H_0 + i\varepsilon} \tag{93}$$

is.

Example 7.3.1 (Electron Scattering): This is an example from the textbook that I was reading which allows for the derivation of the *Advanced* and *Retarded* propagators.

- The transition matrix element is

$$T_{fi} = V_{fi} + \sum_n V_{fn} \frac{1}{E_i - E_n} V_{ni} + \cdots \tag{102}$$

- For electron scattering, the initial and final states are $|i\rangle = |\psi_e^1 \psi_e^2\rangle$ and $\langle f| = \langle \psi_e^3 \psi_e^4|$
- Due to the fact that Coulomb's law is replaced by photon exchange we need a photon field $\phi(x)$.
- This provides

$$V = \frac{1}{2} e \int d^3 x \psi_e(x) \phi(x) \psi_e(x) \tag{103}$$

with the $\frac{1}{2}$ from spin and the e as some coupling constant (which happens to be the electron charge).

Note:
- This interaction turns a $\vec{p}_1$ electron into a $\vec{p}_3$ electron and a $\vec{p}_\gamma$ photon.
- Since both $|i\rangle$ and $|f\rangle$ have 2 electrons and no photons, $V_{ij} = 0$

7.3.1. Retarded and Advanced Propagators

- This gets split into two states. (propagated by the *Retarded* Propagator and the *Advanced* Propagator)

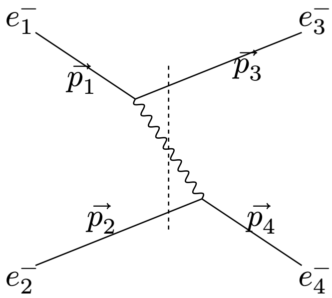

Figure 1: Retarded state

- In the Retarded state, electrons interact via the *Retarded Propagator*.
- We have that

$$|n\rangle = |\psi_e^3 \phi^\gamma \psi_e^2\rangle \tag{104}$$

  - Which implies that

$$\begin{aligned} V_{ni}^R &= \langle \psi_e^3 \phi^\gamma \psi_e^2 | V | \psi_e^1 \psi_e^2 \rangle \\ &= \langle \psi_e^3 \varphi^\gamma | V | \psi_e^1 \rangle \langle \psi_e^2 | \psi_e^2 \rangle \\ &= \langle \psi_e^3 \phi^\gamma | V | \psi_e^1 \rangle \end{aligned} \tag{105}$$

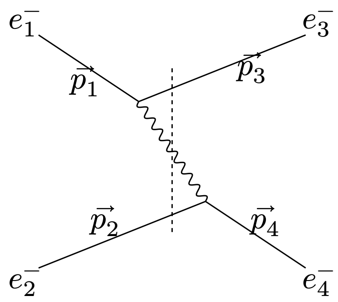

Figure 2: Advanced State

- For the *advanced state*, a similar result is derived
  - $$V_{ni}^A = \langle \psi_e^4 \varphi^\gamma | V | \psi_e^2 \rangle \tag{106}$$

7.3.2. Evaluating The Propagators

- $$V_{ni}^R = \langle \psi_e^3 \varphi^\gamma | V | \psi_e^1 \rangle = \frac{e}{2} \int \mathrm{d}^3 x \langle \psi_e^3 \varphi^\gamma | \psi_e(x) \phi(x) \psi_e(x) | \psi_e^1 \rangle \tag{107}$$

Note:
- Free field (like the photon and electron fields inside of the integral)

$$\langle \phi^\gamma | \phi(x) | 0 \rangle = e^{-i \vec{p} \cdot \vec{x}} \tag{108}$$

- This implies:

$$\begin{aligned} V_{ni}^R &= e \int \mathrm{d}^3 x e^{i(\vec{p}_1 - \vec{p}_2 - \vec{p}_3) \cdot \vec{x}} \\ &= e (2\pi)^3 \, \delta^3(\vec{p}_1 - \vec{p}_2 - \vec{p}_3) \end{aligned} \tag{109}$$

  - This is similar to other matrix elements
- First order:

$$T_{fi} = \int \mathrm{d}^3 \vec{p}_\gamma (2\pi)^3 \, \delta^3(\vec{p}_1 - \vec{p}_2 - \vec{p}_\gamma)(2\pi)^3 \, \delta^3(\vec{p}_2 - \vec{p}_4 + \vec{p}_\gamma) \frac{e^2}{E_i - E_n} \tag{110}$$

  - Energy is not conserved in such intermediate states (otherwise $E_i = E_n$ would blow u

7.3.3. Energy Calculations

- Want to calculate the energy of the states

Denotations

- Momenta of particles are

$$\vec{p}_1, \vec{p}_2, \vec{p}_3, \vec{p}_4, \vec{p}_\gamma \tag{111}$$

8. Feynman Diagrams

Definition 8.1 (Feynman Diagram): A Feynman diagram is something that is used to represent particle interactions, or simplifications of them which are summed. generally to calculate the LSZ reduction for matrix elements. These matrix elements are ultimately used to calculate cross sections as seen in *Cross Sections*.

Definition 8.2 (Feynman Rules): Feynman rules are a set of rules for a theory which is used to evaluate Feynman diagrams.

- A standard set of Feynman rules that can generally apply everywhere are the following

General Position Space Feynman Rules

- At each vertex, a factor of the interaction coupling is given
- All lines are integrated over d^3x after applying the propagator

General Momentum Space Feynman Rules (Arguably much more useful)

- Each vertex of a Feynman Diagram gets a factor of whatever the interaction coupling is.
- Vertices can only occur at points that satisfy interactions in the Lagrangian[8]
- For each edge whose momentum is known (that does not connect to an external vertex), a factor of the propagator for the theory Π is gained
- For each edge whose momentum k is unknown (i.e., a loop), a factor of Π is gained and must be integrated over

$$\frac{\mathrm{d}^4 k}{(2\pi)^4}. \tag{112}$$

- The final product gives $i\mathcal{M}$ for the diagram

Note (Schwinger-Dyson Equation):

[8]For instance, if $\mathcal{L}_{\text{int}} = \frac{g}{3!}\phi^3$, only interactions which involve fields intersecting at three points are included in the set of tree-level diagram

9. Field Theories

Definition 9.1 (Field Theory):

10. QED

Definition 10.1 (Quantum Electrodynamics): Quantum Electrodynamics is a relatively simple theory with only one force involved, mediated by the *photon* and one/two type(s) of fermion depending on what you consider a type of particle (an *electron*/*positron*)

10.1. Scalar QED

- Lagrangian is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^2 + \left|D_\mu \phi\right|^2 - m^2 \phi^* \phi \tag{113}$$

  - $D_\mu \phi = \partial_\mu \phi + ieA_\mu$
  - $\left(D_\mu \phi\right)^* = \partial_\mu \phi^* - iA_u$

10.1.1. Scalar Field Quantization

- Charge associated with $\mathrm{U}(1)$ symmetry

$$\phi \to e^{-i\alpha(x)} \phi \tag{114}$$

Bibliography

[1] M. D. Schwartz, *Quantum Field Theory and the Standard Model*. Cambridge University Press, 2014.